

HU Collection

TMUA Mock Set E Paper 2

Worked Solutions

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Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	C	B	B	C	D	D	C	E	C	D
Q	11	12	13	14	15	16	17	18	19	20
Ans	B	D	E	D	A	F	A	E	F	C

Each solution gives the intended route and names the trap the question was built around. Questions are ordered by increasing difficulty.

Question 1. Answer C.

The converse says: if the diagonals are perpendicular, then the quadrilateral is a rhombus. A counterexample needs perpendicular diagonals but must not be a rhombus. A kite that is not a rhombus is exactly that. Options B and D fail the hypothesis, since their diagonals are not perpendicular, and options A and E are rhombuses, so they satisfy the conclusion.

Question 2. Answer B.

I is true: take $y = x + 1$. II swaps the quantifiers and is false: no single y exceeds every x , since it would have to exceed itself. III fails for negative x . IV is true with $x = 0$. So I and IV. The pair I and II is the whole point: “for every, there is” and “there is, for every” are different claims.

Question 3. Answer B.

If $f(1)$ and $f(3)$ have opposite signs, the continuous function f must cross zero between them: sufficient. It is not necessary, because both roots may lie between 1 and 3: $f(x) = (x - 1.5)(x - 2.5)$ has $f(1)$ and $f(3)$ both positive.

Question 4. Answer C.

I: $2m + 1 = (m + 1)^2 - m^2$. Next, $a^2 - b^2 = (a - b)(a + b)$, and the two factors have the same parity, so the difference is odd or a multiple of 4; it is never $2 \pmod{4}$. Conversely $4m = (m + 1)^2 - (m - 1)^2$. So the neat numbers are exactly those not congruent to $2 \pmod{4}$. II: a product of two such numbers is odd or a multiple of 4. III fails: $1 + 1 = 2$. IV fails: $n = 1$ gives 2. So I and II.

Question 5. Answer D.

The simplification $\frac{x^2 - 4}{x - 2} = x + 2$ is valid only when $x \neq 2$, since the original left-hand side is undefined at $x = 2$. So the root $x = 2$ of the quadratic must be discarded, and $x = 1$ is the only solution. Check: at $x = 1$ both sides equal 3.

Question 6. Answer D.

If $12 \mid n^2$ then $3 \mid n^2$, so $3 \mid n$ (as 3 is prime), and $4 \mid n^2$, so n is even. Hence $6 \mid n$. Nothing more is forced: $n = 6$ has $n^2 = 36$ divisible by 12, and 6 is not divisible by 4 or 12. So $d = 6$. The tempting answer 12 assumes that $12 \mid n^2$ implies $12 \mid n$, which fails because 12 has a square factor.

Question 7. Answer C.

$(x - 1)^2 \geq 0$, so f' has the sign of $x - 3$, except that it is zero at $x = 1$. So f decreases on $(-\infty, 3]$, pausing at $x = 1$ without turning, and increases on $[3, \infty)$. I: the only turning point is the minimum at $x = 3$. II: f decreases from 0 to 2. III is false: $x = 1$ is a stationary point of inflection. IV: f increases after 3.

Question 8. Answer E.

Multiplying an inequality by a number keeps its direction only if that number is positive, so line I needs $x > 0$ and line II needs $y > 0$. The claim is false: $x = 1, y = -2$ gives $1 < 4$. The argument does prove the claim for positive x and y , but that is not what was asserted.

Question 9. Answer C.

Line III is fine for any integers: raise $2^{p/q} = 3$ to the power q . Line IV uses that 2^p is an even integer and 3^q an odd integer, which needs $p \geq 1$ and $q \geq 1$. With negative exponents they are fractions, and “even” and “odd” mean nothing. Line II is what licenses line IV.

Question 10. Answer D.

I: if n is even then n^2 is even; true. II: “ P unless Q ” means “if not Q then P ”, so this reads: if n is divisible by 3 then $9 \mid n^2$; true. III: if n is prime then n is odd; false for $n = 2$. IV: if $n^2 + 1$ is even then n^2 is odd, so n is odd; true. Three are true.

Question 11. Answer B.

$\cos 2 < 0$, since $2 > \frac{\pi}{2}$. $\sin 3 = \sin(\pi - 3)$, which is the sine of a small angle, about 0.14. $\cos 1 = \sin(\frac{\pi}{2} - 1)$, about $\sin 0.57$. $\sin 2 = \sin(\pi - 2)$, about $\sin 1.14$. Sine is increasing on $[0, \frac{\pi}{2}]$ and $1.14 > 1 > 0.57 > 0.14$, so $\sin 2$ is the largest. Reflecting every angle into $[0, \frac{\pi}{2}]$ turns the comparison into one of angles.

Question 12. Answer D.

I: $\sin A = \sin B$ means $A = B$ or $A + B = 180^\circ$, and the second is impossible in a triangle, so $A = B$. II: cosine is one-to-one on $(0^\circ, 180^\circ)$. III: $\tan A = \tan B$ means $A = B$ or they differ by 180° , which is impossible. IV allows $2A + 2B = 180^\circ$, a right angle at C with $A \neq B$: the 3, 4, 5 triangle satisfies it and is not isosceles.

Question 13. Answer E.

N is odd, so statements (1) and (2) are false, and both (3) and (4) must be true: N is an odd perfect square greater than 40 and at most 50. That is 49. Option C, 36, satisfies (1), (2) and (3) but is even.

Question 14. Answer D.

Compute: $a = 2$ gives $2, 2, \dots$; $a = 0$ gives $0, -2, 2, 2, \dots$; $a = 1$ gives $1, -1, -1, \dots$; $a = -2$ gives $-2, 2, 2, \dots$. All four are bounded. $a = 3$ gives $3, 7, 47, \dots$ and $a = -2.5$ gives $4.25, \dots$, both unbounded, since once $|x_n| > 2$ we have $x_{n+1} - 2 = x_n^2 - 4 > 0$ and the excess over 2 grows. So four. The rule is that the sequence is bounded exactly when $|a| \leq 2$.

Question 15. Answer A.

For an integer n , $p(n) - p(0)$ is a sum of multiples of n , and $p(n) - p(1)$ is a sum of terms $a_k(n^k - 1)$, each a multiple of $n - 1$. For even n , $p(n)$ has the parity of $p(0)$; for odd n , that of $p(1)$. Either way $p(n)$ is odd, never 0: I and III hold. II and IV fail for $p(x) = 2x - 1$, with $p(0) = -1$ and $p(1) = 1$, whose root is $\frac{1}{2}$.

Question 16. Answer F.

II: put $x = \frac{1}{2}$. Reflecting $[0, 1]$ in $x = \frac{1}{2}$ shows that $\int_0^1 f(1-x) dx = \int_0^1 f(x) dx$, so adding gives $2 \int_0^1 f = \int_0^1 1 = 1$, and I holds. For IV, the same reflection turns $\int_0^1 xf(x) dx$ into $\int_0^1 (1-x)f(1-x) dx = \int_0^1 (1-x)(1-f(x)) dx$. III fails: $f(x) = \frac{1}{2} + \sin(2\pi(x - \frac{1}{2}))$ satisfies the condition but is not increasing.

Question 17. Answer A.

Opposite angles of a cyclic quadrilateral sum to 180° . I: a parallelogram's opposite angles are equal, so each is 90° . II follows from I, since a rectangle that is a rhombus is a square. III: the angles on each parallel side are supplementary to their neighbours and so equal in pairs, which makes the trapezium isosceles. IV: a kite has one pair of equal opposite angles, so each is 90° . V fails: many cyclic quadrilaterals have perpendicular diagonals without two pairs of equal adjacent sides.

Question 18. Answer E.

I checks five cases and says nothing about $n = 6$ or beyond, so it is not a proof. II covers every integer through its remainder, since $n^2 + n + 1$ modulo 5 depends only on n modulo 5: a complete proof. III is also complete: the last digit of $n(n+1)$ depends only on the last digit of n , and the ten cases give only 0, 2 and 6. A last digit of 1, 3 or 7 rules out divisibility by 5.

Question 19. Answer F.

$S_n = An^2 + Bn$ with $A \neq 0$. $S_{10} = S_{20}$ gives $300A + 10B = 0$, so $S_n = A(n^2 - 30n)$. I: $S_{30} = 0$. IV: $n^2 - 30n$ is least at $n = 15$, so S_{15} is the extreme value. II: $u_{11} + \dots + u_{20} = S_{20} - S_{10} = 0$, and these ten terms pair up as five pairs each equal to $u_{15} + u_{16}$. III fails: in $u_n = 31 - 2n$ we have $S_{10} = S_{20} = 200$ but $u_{15} = 1$.

Question 20. Answer C.

IV: for a base between 0 and 1, a larger exponent gives a smaller power, and $b > a$. I needs no calculus: $a^b < b^b$, because raising to the positive power b preserves $a < b$, and $b^b < b^a$ by the same fact as IV with base b . So $a^b < b^b < b^a$. III: $\log_b a = \frac{\ln a}{\ln b}$ is a ratio of two negative numbers, and $|\ln a| > |\ln b|$, so it exceeds 1. II fails: $a = 0.1$, $b = 0.3$ gives $a^a \approx 0.794$ and $b^b \approx 0.697$. The function x^x decreases before it increases, so no ordering is forced.